Physics Unit 2 · Practice Quiz 1

Static and kinetic friction · inclined planes · universal gravitation · circular motion · Newton's laws

Name: _____ Date: _____ Period: _____

Instructions. Part A: 20 multiple-choice questions, 2 points each (40 points). Part B: 4 free-response questions, 10 points each (40 points). Total 80 points. Suggested time: 50 minutes.

Reference. $g = 9.81$ m/s² · $G = 6.67 \times 10^{-11}$ N·m²/kg² · $M_E = 5.97 \times 10^{24}$ kg · $R_E = 6.38 \times 10^{6}$ m. Give answers to 3 significant figures with units. Show your work on the free-response questions. A calculator may be used (degree mode).

Part A · Multiple choice (2 points each)

1. You push harder and harder on a heavy crate, but it still does not move. The static friction force on the crate

A increases to match your push
B decreases
C is zero because the crate is not moving
D stays the same

2. A 20 kg box sits on a level floor ($\mu_s = 0.50$). You push it horizontally with 60 N and it does not move. What is the friction force on the box?

A 38.1 N **B** 0 N
C 98.1 N **D** 60 N

3. A 20 kg crate rests on a level floor with $\mu_s = 0.45$. What is the largest horizontal push that will NOT move it?

A 9.00 N **B** 196 N
C 436 N **D** 88.3 N

4. A crate is pushed across a floor at constant velocity. The push is

A equal to the crate's weight
B equal to the kinetic friction force
C less than the kinetic friction force
D greater than the kinetic friction force

5. A 100 N horizontal push slides a 20 kg crate across a floor with $\mu_k = 0.30$. What is its acceleration?

A 4.70 m/s² **B** 5.00 m/s²
C 2.06 m/s² **D** 7.94 m/s²

6. A puck slides across a level surface at 12 m/s with $\mu_k = 0.10$ and no push. How far does it slide before it stops?

A 147 m **B** 73.4 m
C 12.2 m **D** 7.34 m

7. On an incline at angle θ, the component of the weight that pulls the object down along the slope is

A mg
B mg sin θ
C mg cos θ
D mg tan θ

8. What is the normal force on a 60 kg block resting on a 25° incline?

A 649 N
B 589 N
C 249 N
D 533 N

9. A block slides down a frictionless 30° incline. What is its acceleration?

A 4.90 m/s²
B 8.50 m/s²
C 5.66 m/s²
D 9.81 m/s²

10. A 10 kg box slides down a 35° ramp with $\mu_k = 0.30$. What is its acceleration?

A 3.22 m/s²
B 8.04 m/s²
C 6.35 m/s²
D 5.63 m/s²

11. Compare the force Earth exerts on the Moon with the force the Moon exerts on Earth.

A Earth's pull is about 81 times larger
B They are equal in size and opposite in direction
C The Moon's pull is larger
D The Moon exerts no force on Earth

12. Two objects attract each other with a gravitational force F. The distance between their centers is doubled. What is the new force?

A F/2
B 4F
C 2F
D F/4

13. Two students, 70 kg and 60 kg, sit 1.5 m apart (center to center). What is the gravitational force between them? ($G = 6.67 \times 10^{-11}$ N·m²/kg²)

A 1.25×10^{-7} N
B 1.87×10^{-7} N
C 1.87×10^{3} N
D 3.11×10^{-8} N

14. A planet has 2 times Earth's mass and 2 times Earth's radius. What is g at its surface?

A 78.5 m/s²
B 19.6 m/s²
C 4.91 m/s²
D 9.81 m/s²

15. An object moves in a circle at constant speed. The direction of its acceleration is

A toward the center of the circle
B zero; it is not accelerating
C away from the center
D along the direction of motion (tangent)

16. A car rounds a flat (unbanked) curve at constant speed. What force provides the centripetal force?

A Kinetic friction
B The normal force
C Static friction from the road on the tires
D The engine

17. A runner moves at 12 m/s around a circle of radius 30 m. What is its centripetal acceleration?

A 0.800 m/s² **B** 4,320 m/s²
C 0.400 m/s² **D** 4.80 m/s²

18. What is the maximum speed at which a car can round a flat curve of radius 40 m if μ_s = 0.80?

A 22.1 m/s **B** 17.7 m/s
C 5.66 m/s **D** 314 m/s

19. You stand on a scale in an elevator that is accelerating upward. The scale reads

A zero **B** less than your weight
C more than your weight **D** exactly your weight

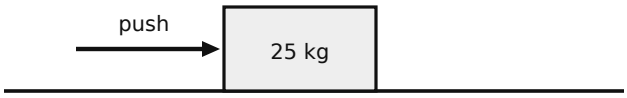
20. Newton's third law force pairs

A act on the same object and cancel
B are always a weight and a normal force
C act on two different objects and are equal and opposite
D are only equal when nothing accelerates

Part B · Free response (10 points each)

21. Static and kinetic friction (10 points)

A 25 kg crate sits on a level floor. μ_s = 0.40 and μ_k = 0.30.



(a) A 60 N horizontal push does not move the crate. Draw and label its free-body diagram. **[2]**

(b) What is the friction force on the crate while it is pushed with 60 N? Explain. **[2]**

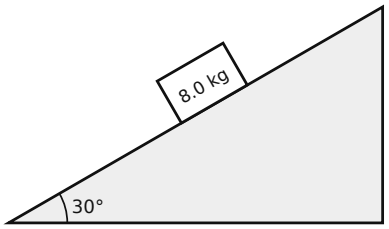
(c) What is the smallest horizontal push that will start the crate moving? **[2]**

(d) The crate is pushed with 120 N. Find its acceleration. **[2]**

(e) While the crate slides at 3.0 m/s the push is removed. How far does it slide before it stops? **[2]**

22. Inclined plane (10 points)

An 8.0 kg block is released from rest on a 30° incline. $\mu_k = 0.20$.



(a) Draw and label the free-body diagram of the block as it slides down. **[2]**

(b) Find the components of the weight parallel and perpendicular to the incline. **[2]**

(c) Find the normal force and the kinetic friction force. **[2]**

(d) Find the acceleration of the block. **[2]**

(e) How fast is the block moving after it has slid 3.0 m? **[2]**

23. Universal gravitation and orbits (10 points)

A 500 kg satellite moves in a circular orbit 600 km above Earth's surface.

(a) What is the radius of its orbit, r (distance from Earth's center)? **[1]**

(b) Find the gravitational force Earth exerts on the satellite. **[3]**

(c) What is the gravitational field strength g at that height? **[2]**

(d) Find the satellite's orbital speed. **[2]**

(e) Find its orbital period in minutes. **[2]**

24. Circular motion on a flat curve (10 points)

A 1200 kg car drives around a flat (unbanked) curve of radius 45 m. μ_s between the tires and the road is 0.70.

(a) Which force provides the centripetal force, and in which direction does it point? **[2]**

(b) Find the maximum speed at which the car can take the curve without skidding. **[3]**

(c) The car takes the curve at 15 m/s. How large is the friction force on the car? **[2]**

(d) The driver doubles the speed to 30 m/s. What friction force would be needed now? What happens to the car, and in which direction does it move? **[3]**